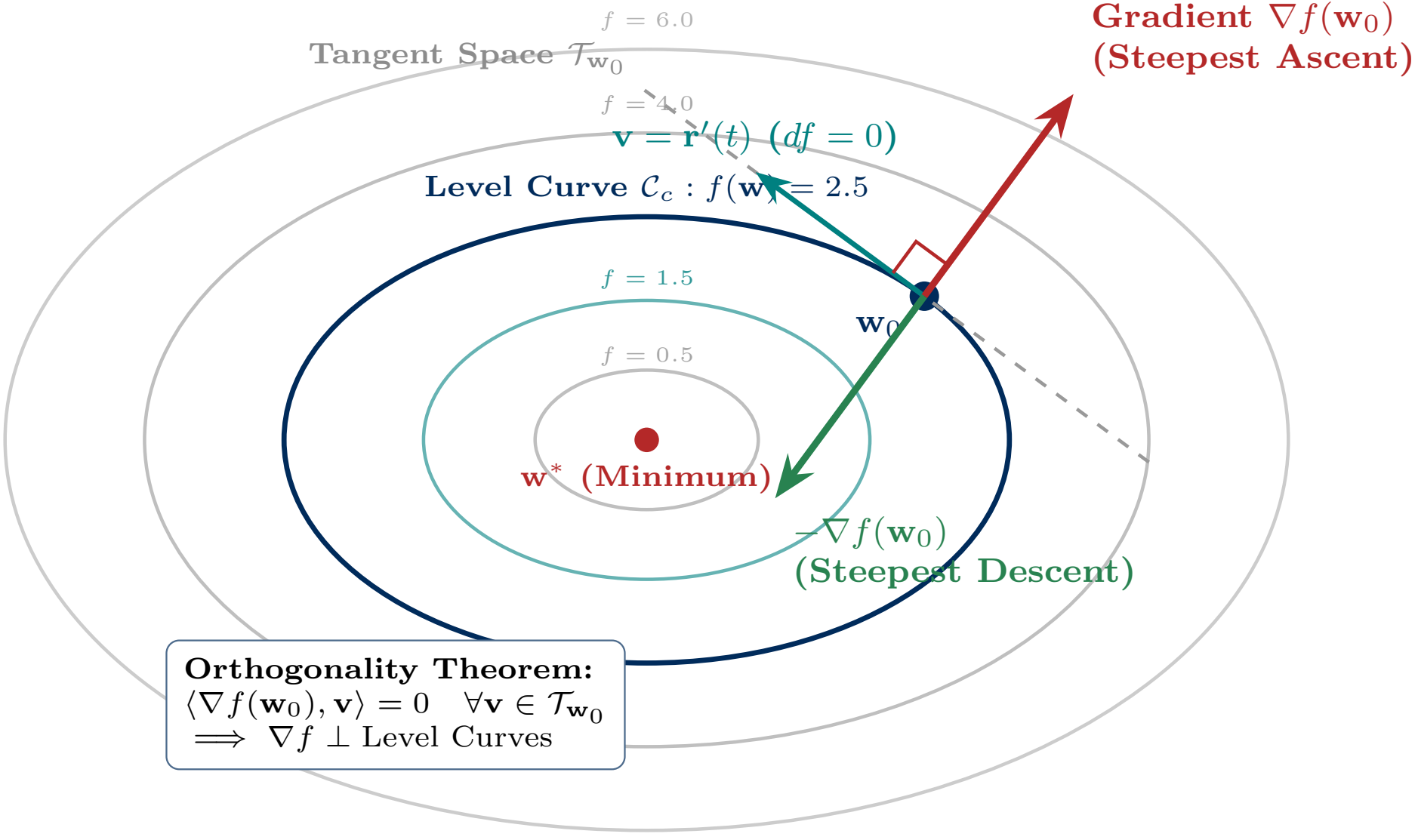

$f = 6.0$
Tangent Space $\mathcal{T}_{\mathbf{w}_0}$
$f = 4.0$
$\mathbf{v} = \mathbf{r}'(t)$ $(df = 0)$
Level Curve $\mathcal{C}_c : f(\mathbf{w}) = 2.5$
$f = 1.5$
$f = 0.5$
$\mathbf{w}_0$
$\mathbf{w}^*$ (Minimum)
Gradient $\nabla f(\mathbf{w}_0)$
(Steepest Ascent)
$-\nabla f(\mathbf{w}_0)$
(Steepest Descent)
Orthogonality Theorem:
$\langle \nabla f(\mathbf{w}_0), \mathbf{v} \rangle = 0 \quad \forall \mathbf{v} \in \mathcal{T}_{\mathbf{w}_0}$
$\implies \nabla f \perp$ Level Curves